2010 Team Answers

T-1. $(-6, 6)$ and $(2, 6)$ [must have both answers, in either order]

T-2. 9

T-3. 100337

T-4. 392

T-5. 84

T-6. 3851

T-7. $\dfrac{\sqrt{2}}{3}$

T-8. 529

T-9. $5\sqrt[3]{4}$

T-10. $\dfrac{375}{182}$

2010 Team Solutions

T-1. Rearrange the terms in the first equation to yield $x^2 + 4x + 12 = 6y - y^2 + 24$, so that the two equations together yield $4y = 6y - y^2 + 24$, or $y^2 - 2y - 24 = 0$, from which $y = 6$ or $y = -4$. If $y = 6$, then $x^2 + 4x + 12 = 24$, from which $x = -6$ or $x = 2$. If $y = -4$, then $x^2 + 4x + 12 = -16$, which has no real solutions because $x^2 + 4x + 12 = (x + 2)^2 + 8 \geq 8$ for all real x. So there are two ordered pairs satisfying the system, namely $\mathbf{(-6, 6)}$ and $\mathbf{(2, 6)}$.

T-2. If $\log^*(a) = 3$, then $\log(\log(\log(a))) \leq 1$ and $\log(\log(a)) > 1$. If $\log(\log(a)) > 1$, then $\log(a) > 10$ and $a > 10^{10}$. Because the problem asks for the smallest such a that is an integer, choose $a = 10^{10} + 1 = 10{,}000{,}000{,}001$, which has **9** zeros.

T-3. If a number N has k base 10 digits, then its maximum point value is $(k-1)+(k-2)+\cdots+1 = \frac{1}{2}(k-1)(k)$. So if $k \leq 5$, the number N is worth at most 10 points. Therefore the desired number has at least six digits. If $100{,}000 < N < 101{,}000$, then N is of the form $100\underline{A}\,\underline{B}\,\underline{C}$, which could yield 12 possible primes, namely $1\underline{A}$, $1\underline{B}$, $1\underline{C}$, $0\underline{A}$ (twice), $0\underline{B}$ (twice), $0\underline{C}$ (twice), $\underline{A}\,\underline{B}$, $\underline{A}\,\underline{C}$, $\underline{B}\,\underline{C}$. So search for N of the form $100\underline{A}\,\underline{B}\,\underline{C}$, starting with lowest values first. Notice that if any of A, B, or C is not a prime, at least two points are lost, and so all three numbers must be prime. Proceed by cases:

First consider the case $A = 2$. Then $1\underline{A}$ is composite, so all of $1\underline{B}$, $1\underline{C}$, $\underline{A}\,\underline{B}$, $\underline{A}\,\underline{C}$, $\underline{B}\,\underline{C}$ must be prime. Considering now the values of $1\underline{B}$ and $1\underline{C}$, both B and C must be in the set $\{3, 7\}$. Because 27 is composite, $B = C = 3$, but then $\underline{B}\,\underline{C} = 33$ is composite. So A cannot equal 2.

If $A = 3$, then $B \neq 2$ because both 12 and 32 are composite. If $B = 3$, $1\underline{B}$ is prime but $\underline{A}\,\underline{B} = 33$ is composite, so all of C, $1\underline{C}$, and $3\underline{C}$ must be prime. These conditions are satisfied by $C = 7$ and no other value. So $A = B = 3$ and $C = 7$, yielding $N = \mathbf{100337}$.

T-4. Only two triangles have no sides that are sides of the original hexagon: $A_1A_3A_5$ and $A_2A_4A_6$. For each of these triangles, there are $2^3 - 1 = 7$ colorings in which at least one side is red, for a total of $7 \cdot 7 = 49$ colorings of those six diagonals. The colorings of the three central diagonals $\overline{A_1A_4}, \overline{A_2A_5}, \overline{A_3A_6}$ are irrelevant because the only triangles they can form include sides of the original hexagon, so they can be colored in $2^3 = 8$ ways, for a total of $8 \cdot 49 = \mathbf{392}$ colorings.

T-5. Let k denote the number of distinct prime divisors of n, so that $n = p_1^{a_1}p_2^{a_2}\cdots p_k^{a_k}$, $a_i > 0$. Then if $d(x)$ denotes the number of positive divisors of x,

$$d\left(n^n\right) = (a_1n + 1)\,(a_2n + 1)\cdots(a_kn + 1) \geq (n + 1)^k. \qquad (*)$$

Note that if $n \geq 99$ and $k \geq 3$, then $d(n^n) \geq 100^3 = 10^6$, so $102 = 2 \cdot 3 \cdot 17$ is an upper bound for the solution. Look for values less than 99, using two observations: (1) all $a_i \leq 6$

(because $p^7 > 99$ for all primes); and (2) $k \leq 3$ (because $2 \cdot 3 \cdot 5 \cdot 7 > 99$). These two facts rule out the cases $k = 1$ (because $(*)$ yields $d \leq (6n+1)^1 < 601$) and $k = 2$ (because $d(n^n) \leq (6n+1)^2 < 601^2$).

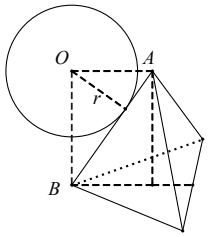
So $k = 3$. Note that if $a_1 = a_2 = a_3 = 1$, then from $(*)$, $d(n^n) = (n+1)^3 < 10^6$. So consider only $n < 99$ with exactly three prime divisors, and for which not all exponents are 1. The only candidates are 60, 84, and 90; of these, $n = 84$ is the smallest one that works:

$$\begin{aligned} d(60^{60}) &= d(2^{120} \cdot 3^{60} \cdot 5^{60}) = 121 \cdot 61 \cdot 61 < 125 \cdot 80 \cdot 80 = 800{,}000; \\ d(84^{84}) &= d(2^{168} \cdot 3^{84} \cdot 7^{84}) = 169 \cdot 85 \cdot 85 > 160 \cdot 80 \cdot 80 = 1{,}024{,}000. \end{aligned}$$

Therefore $n = \mathbf{84}$ is the least positive integer n such that $d(n^n) > 1{,}000{,}000$.

T-6. Divide into cases according to the number of E's in the word. If there are only two E's, then the word must have two non-E letters, represented by ?'s. There are $\binom{4}{2} = 6$ arrangements of two E's and two ?'s, and each of the ?'s can be any of 25 letters, so there are $6 \cdot 25^2 = 3750$ possible words. If there are three E's, then the word has exactly one non-E letter, and so there are 4 arrangements times 25 choices for the letter, or 100 possible words. There is one word with four E's, hence a total of **3851** words.

T-7. Let O be the center of the sphere, A be the top vertex of one tetrahedron, and B be the center of the hexagon.



Then BO equals the height of the tetrahedron, which is $\frac{\sqrt{6}}{3}$. Because A is directly above the centroid of the bottom face, AO is two-thirds the length of the median of one triangular face, so $AO = \frac{2}{3}\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{3}$. The radius of the sphere is the altitude to hypotenuse $\overline{AB}$ of $\triangle ABO$, so the area of $\triangle ABO$ can be represented in two ways: $[ABO] = \frac{1}{2}AO \cdot BO = \frac{1}{2}AB \cdot r$. Substitute given and computed values to obtain $\frac{1}{2}\left(\frac{\sqrt{3}}{3}\right)\left(\frac{\sqrt{6}}{3}\right) = \frac{1}{2}(1)(r)$, from which $r = \frac{\sqrt{18}}{9} = \frac{\sqrt{\mathbf{2}}}{\mathbf{3}}$.

T-8. Divide into cases according to the number of right turns Derek makes.

- There is one route involving only one turn: move first to $(0, 1)$ and then to $(1, 1)$.
- If he makes two turns, he could move up to $(0, a)$ then to $(1, a)$ and then down to $(1, 1)$. In order to do this, a must satisfy $1 < a \leq 17$, leading to 16 options.

- If Derek makes three turns, his path is entirely determined by the point at which he turns for the second time. If the coordinates of this second turn point are (a, b), then both a and b are between 2 and 17 inclusive, yielding $(17-1)^2$ possibilities.
- If Derek makes four turns, his last turn must be from facing in the $-x$-direction to the $+y$-direction. For this to be his last turn, it must occur at $(1, 0)$. Then his next-to-last turn could be at any $(a, 0)$, with $1 < a \leq 17$, depending on the location of his second turn as in the previous case. This adds another $(17-1)^2$ possibilities.
- It is impossible for Derek to make more than four turns and get to $(1, 1)$ without crossing or overlapping his path.

Summing up the possibilities gives $1 + 16 + 16^2 + 16^2 = \mathbf{529}$ possibilities.

T-9. Let the roots of the first equation be p, q, r and the roots of the second equation be p, q, s. Then $pqr = -10$ and $pqs = -50$, so $\frac{s}{r} = 5$. Also $p + q + r = 0$ and $p + q + s = -B$, so $r - s = B$. Substituting yields $r - 5r = -4r = B$, so $r = -\frac{B}{4}$ and $s = -\frac{5B}{4}$. From the second given equation, $pq + ps + qs = pq + s(p + q) = 0$, so $pq - \frac{5B}{4}(p + q) = 0$, or $pq = \frac{5B}{4}(p + q)$. Because $p + q + r = 0$, $p + q = -r = \frac{B}{4}$, and so $pq = \frac{5B^2}{16}$. Because $pqr = -10$ and $r = -\frac{B}{4}$, conclude that $pq = \frac{40}{B}$. Thus $\frac{5B^2}{16} = \frac{40}{B}$, so $B^3 = 128$ and $B = 4\sqrt[3]{2}$. Then $pq = \frac{5B^2}{16}$ implies that $pq = \mathbf{5\sqrt[3]{4}}$ (and $r = -\sqrt[3]{2}$).

Alternate Solution: Let the common roots be p and q. Then the following polynomials (linear combinations of the originals) must also have p and q as common zeros:

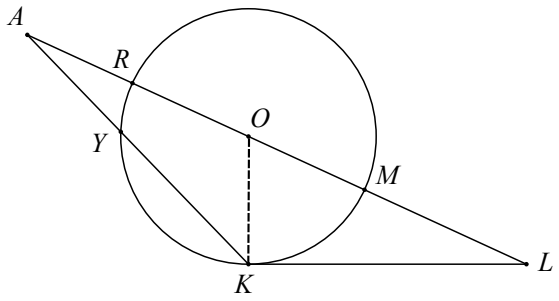
$$\begin{aligned}(x^3 + Bx^2 + 50) - (x^3 + Ax + 10) &= Bx^2 - Ax + 40\\ -(x^3 + Bx^2 + 50) + 5(x^3 + Ax + 10) &= 4x^3 - Bx^2 + 5Ax.\end{aligned}$$

Because $pq \neq 0$, neither p nor q is zero, so the second polynomial has zeros p, q, and 0. Therefore p and q are zeros of $4x^2 - Bx + 5A$. [This result can also be obtained by using the Euclidean Algorithm on the original polynomials.]

Because the two quadratic equations have the same zeros, their coefficients are proportional: $\frac{4}{B} = \frac{5A}{40} \Rightarrow AB = 32$ and $\frac{4}{B} = \frac{-B}{-A} \Rightarrow 4A = B^2$. Hence $\frac{128}{B} = B^2$ and $B^3 = 128$, so $B = 4\sqrt[3]{2}$. Rewriting the first quadratic as $B\left(x^2 - \frac{A}{B}x + \frac{40}{B}\right)$ shows that the product $pq = \frac{40}{B} = \mathbf{5\sqrt[3]{4}}$.

Alternate Solution: Using the sum of roots formulas, notice that $pq+ps+qs = p+q+r = 0$. Therefore $0 = pq + ps + qs - (p + q + r)s = pq - rs$, and $pq = rs$. Hence $(pq)^3 = (pqr)(pqs) = (-10)(-50) = 500$, so $pq = \mathbf{5\sqrt[3]{4}}$.

T-10. Notice that $\overline{OK} \perp \overline{KL}$, and let r be the radius of ω.



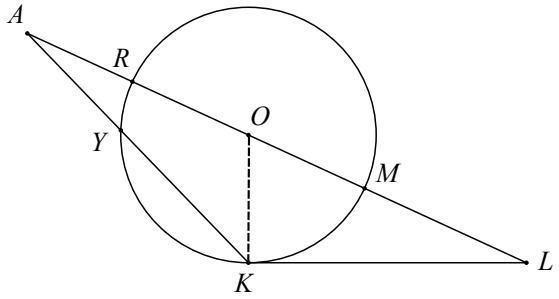
Then consider right triangle OKL. Because $ML = 2$, $OK = r$, and $OL = r + 2$, it follows that $r^2 + 3^2 = (r + 2)^2$, from which $r = \frac{5}{4}$.

Because $\text{m}\angle YKL = \frac{1}{2}\text{m}\widehat{YRK}$ and $\text{m}\angle YMK = \frac{1}{2}\text{m}\widehat{YK}$, it follows that $\text{m}\angle YKL + \text{m}\angle YMK = 180°$. By the given condition, $\text{m}\angle YKL - \text{m}\angle YMK = 90°$. It follows that $\text{m}\angle YMK = 45°$ and $\text{m}\angle YKL = 135°$. hence $\text{m}\widehat{YK} = 90°$. Thus,

$$\overline{YO} \perp \overline{OK} \quad \text{and} \quad \overline{YO} \parallel \overline{KL}. \tag{$*$}$$

From here there are several solutions:

First Solution: Compute $[AKM]$ as $\frac{1}{2}\text{base} \cdot \text{height}$, using base $\overline{AM}$.



Because of $(*)$, $\triangle AYO \sim \triangle AKL$. To compute AM, notice that in $\triangle AYO$, $AO = AM - r$, while in $\triangle AKL$, the corresponding side $AL = AM + ML = AM + 2$. Therefore:

$$\begin{aligned} \frac{AO}{AL} &= \frac{YO}{KL} \\ \frac{AM - \frac{5}{4}}{AM + 2} &= \frac{5/4}{3}, \end{aligned}$$

from which $AM = \frac{25}{7}$. Draw the altitude of $\triangle AKM$ from vertex K, and let h be its length. In right triangle OKL, h is the altitude to the hypotenuse, so $\frac{h}{3} = \sin(\angle KLO) = \frac{r}{r+2}$. Hence $h = \frac{15}{13}$. Therefore $[AKM] = \frac{1}{2} \cdot \frac{25}{7} \cdot \frac{15}{13} = \mathbf{\frac{375}{182}}$.

Second Solution: By the Power of the Point Theorem, $LK^2 = LM \cdot LR$, so

$$LR = \frac{9}{2},$$
$$RM = LR - LM = \frac{5}{2},$$
$$OL = r + ML = \frac{13}{4}. \quad (\dagger)$$

From $(*)$, we know that $\triangle AYO \sim \triangle AKL$. Hence by $(\dagger)$,

$$\frac{AL}{AO} = \frac{AL}{AL - OL} = \frac{KL}{YO} = \frac{3}{5/4} = \frac{12}{5}, \quad \text{thus} \quad AL = \frac{12}{7} \cdot OL = \frac{12}{7} \cdot \frac{13}{4} = \frac{39}{7}.$$

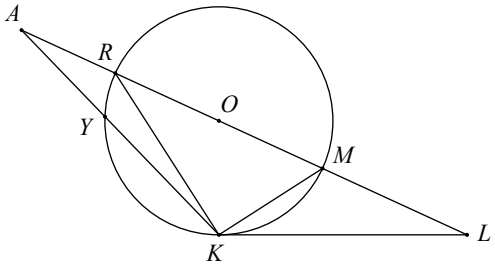
Hence $AM = AL - 2 = \frac{25}{7}$. The ratio between the areas of triangles AKM and RKM is equal to

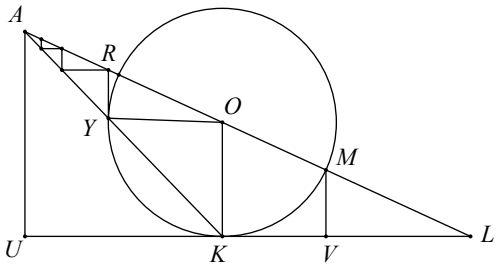
$$\frac{[AKM]}{[RKM]} = \frac{AM}{RM} = \frac{25/7}{5/2} = \frac{10}{7}.$$

Thus $[AKM] = \frac{10}{7} \cdot [RKM]$.

Because $\angle KRL$ and $\angle MKL$ both subtend $\widehat{KM}$, $\triangle KRL \sim \triangle MKL$. Therefore $\frac{KR}{MK} = \frac{LK}{LM} = \frac{3}{2}$. Thus let $KR = 3x$ and $MK = 2x$ for some positive real number x. Because RM is a diameter of ω (see left diagram below), $\mathrm{m}\angle RKM = 90°$. Thus triangle RKM is a right triangle with hypotenuse $\overline{RM}$. In particular, $13x^2 = KR^2 + MK^2 = RM^2 = \frac{25}{4}$, so $x^2 = \frac{25}{52}$ and $[RKM] = \frac{RK \cdot KM}{2} = 3x^2$. Therefore

$$[AKM] = \frac{10}{7} \cdot [RKM] = \frac{10}{7} \cdot 3 \cdot \frac{25}{52} = \mathbf{\frac{375}{182}}.$$





Third Solution: Let U and V be the respective feet of the perpendiculars dropped from A and M to $\overleftrightarrow{KL}$. From $(*)$, $\triangle AKL$ can be dissected into two infinite progressions of triangles: one progression of triangles similar to $\triangle OKL$ and the other similar to $\triangle YOK$, as shown in the right diagram above. In both progressions, the corresponding sides of the triangles have common ratio equal to

$$\frac{YO}{KL} = \frac{5/4}{3} = \frac{5}{12}.$$

Thus

$$AU = \frac{5}{4}\left(1 + \frac{5}{12} + \left(\frac{5}{12}\right)^2 + \cdots\right) = \frac{5}{4} \cdot \frac{12}{7} = \frac{15}{7}.$$

Because $\triangle LMV \sim \triangle LOK$, and because $LO = \frac{13}{4}$ by (†),

$$\frac{MV}{OK} = \frac{LM}{LO}, \quad \text{thus} \quad MV = \frac{OK \cdot LM}{LO} = \frac{\frac{5}{4} \cdot 2}{\frac{13}{4}} = \frac{10}{13}.$$

Finally, note that $[AKM] = [AKL] - [KLM]$. Because $\triangle AKL$ and $\triangle KLM$ share base $\overline{KL}$,

$$[AKM] = \frac{1}{2} \cdot 3 \cdot \left(\frac{15}{7} - \frac{10}{13}\right) = \boxed{\frac{\mathbf{375}}{\mathbf{182}}}.$$